PROBLEM 09

Artificial Gravity in a Rotating Cylinder

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Centrifugal + Coriolis Forces

In "2001: A Space Odyssey" an idea to obtain artificial gravity in a spaceship was presented. It consists of a cylinder rotating at constant angular velocity around its axis. The astronauts would live in the inner surface of the cylinder and the centrifugal force would act as artificial gravity.

(a) If the cylinder has radius R, determine the angular velocity ω required to obtain a given level artificial gravity g. Consider $R = 10$ m and $g = 9.81$ m/s^2 for a particular case.

(b) An astronaut runs inside the surface of the cylinder with a constant speed u with respect to the cylinder. Determine the artificial gravity that she feels (neglect the height of the astronaut h with respect to the radius of the cylinder R, $h \ll R$).

(c) What is the condition for her to feel no artificial gravity at all?

Worked Solution

Reference Frames, Bases and Coordinates

- **Physical system:** a cylindrical hull of radius R spins at constant rate ω about its axis in deep space. An astronaut P of mass m stands on, or runs along, the inner surface.
- **Why the hull frame.** The astronaut lives in the hull: the floor, her running speed u and what she "feels" are all defined with respect to the cylinder. In the frame S_1 attached to the hull the floor is fixed, and the only price is to add the inertia forces of Notes Eq. 4.62. The centrifugal force is exactly the "artificial gravity" of the statement; the Coriolis force is what changes when she runs.
- **Hypotheses:**

1. No external gravity (deep space); the only actual force on the astronaut is the total force $\mathbf{N}$ of the floor on her feet (normal reaction plus any friction; part (b) shows that, for uniform running, $\mathbf{N}$ comes out purely radial).
2. $h \ll R$: she is a point particle on the surface, at distance R from the axis.
3. ω constant; the axis point O is fixed in an inertial frame S_0.

- **Artificial gravity felt.** A body feels as "weight" the force it has to receive from the floor. We define the artificial gravity as $g_{\text{art}} = |\mathbf{N}|/m$, the reading per unit mass of a scale under her feet, directed opposite to $\mathbf{N}$ (i.e. away from the axis, "down" for her).
- **Fixed (inertial) frame:** $S_0 = \{O; \mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$, with O on the axis and $\mathbf{k}_0$ along it.
- **Moving frame:** $S_1 = \{O; \mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1\}$ fixed to the hull, $\mathbf{k}_1 = \mathbf{k}_0$. The origin of S_1 is the fixed point O; S_1 coincides with S_0 at $t = 0$.
- **Local cylindrical basis of the astronaut (in S_1):** $\{\mathbf{e}_r, \mathbf{e}_\theta, \mathbf{k}_1\}$, radial (from the axis to the floor), circumferential (in the sense of rotation) and axial; deduced in the next section.
- **Coordinates and degrees of freedom:** on the cylinder surface, 2 coordinates: azimuth φ in S_1 (measured from $\mathbf{i}_1$ in the sense of rotation) and axial position z. Running at constant speed u means $u_\theta = R\dot{\varphi}$ and $u_z = \dot{z}$ constant with $u_\theta^2 + u_z^2 = u^2$.

Symbol	Meaning	SI unit
R	radius of the cylinder (10 m in the numerical case)	m
ω	spin rate of the cylinder	rad/s
g	target artificial gravity (9.81 m/s^2)	m/s^2
u	running speed relative to the cylinder	m/s
u_θ, u_z	circumferential (positive in the sense of rotation) and axial components of the running velocity	m/s
m	mass of the astronaut	kg

Change of Basis and Position Vector

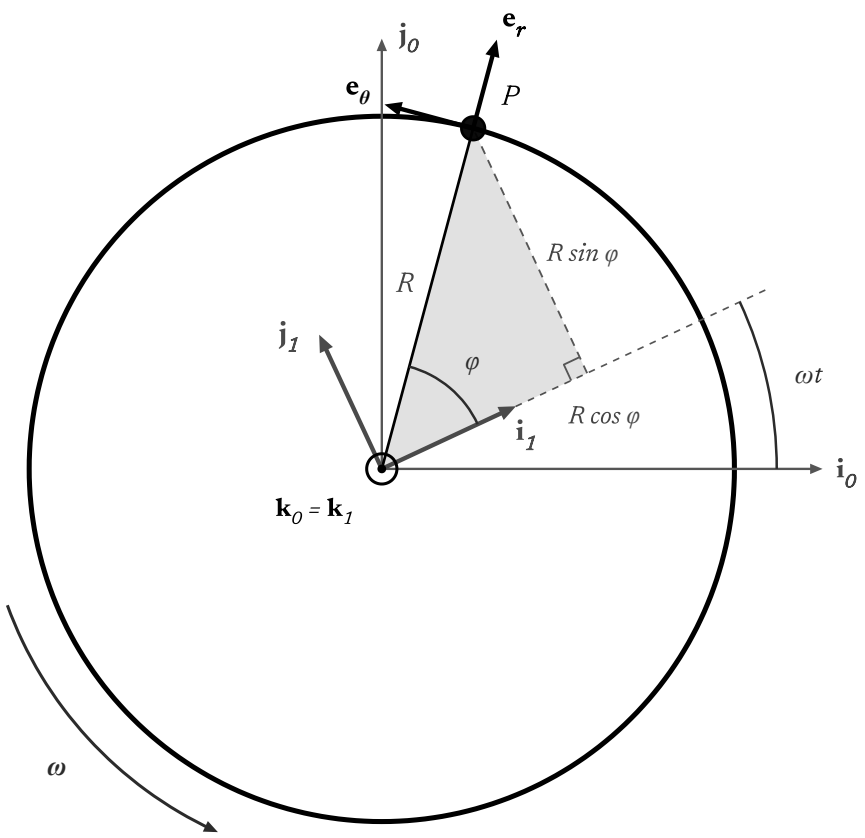


Figure 1. Cross-section along the axis ($\mathbf{k}_0 = \mathbf{k}_1$ out of the paper, $\odot$). The hull axes $\mathbf{i}_1, \mathbf{j}_1$ are $\mathbf{i}_0, \mathbf{j}_0$ turned counter-clockwise by ωt. In the shaded right triangle the hypotenuse is R, the leg along $\mathbf{i}_1$ is $R\cos\varphi$ and the leg along $\mathbf{j}_1$ is $R\sin\varphi$; $\mathbf{e}_r, \mathbf{e}_\theta$ in orange.

The basis $\mathcal{B}_1$. The hull turns about its axis $\mathbf{k}_0$, so the cross-section is drawn looking along the axis, with $\mathbf{k}_0 = \mathbf{k}_1$ pointing out of the paper, towards the reader ($\odot$). With $\mathbf{i}_0$ to the right, the right-hand rule $\mathbf{i}_0 \times \mathbf{j}_0 = \mathbf{k}_0$ puts $\mathbf{j}_0$ up the page, and a positive ω turns the hull counter-clockwise. After a time t, $\mathbf{i}_1$ has turned the angle ωt from $\mathbf{i}_0$:

- $\mathbf{i}_1$ is the hypotenuse (length 1) of a right triangle with the leg $\cos\omega t$ along $\mathbf{i}_0$ (adjacent) and the leg $\sin\omega t$ along $+\mathbf{j}_0$ (opposite): $\mathbf{i}_1 = \cos\omega t\,\mathbf{i}_0 + \sin\omega t\,\mathbf{j}_0$.
- $\mathbf{j}_1$ is $\mathbf{j}_0$ turned the same angle counter-clockwise; turning from "up" towards the left, its opposite leg $\sin\omega t$ points along $-\mathbf{i}_0$, which gives the minus sign: $\mathbf{j}_1 = -\sin\omega t\,\mathbf{i}_0 + \cos\omega t\,\mathbf{j}_0$.
- $\mathbf{k}_1 = \mathbf{k}_0$, the axis of rotation.

$$[{}_0R_1] = \begin{pmatrix} \cos\omega t & -\sin\omega t & 0 \\ \sin\omega t & \cos\omega t & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \det[{}_0R_1] = \cos\omega t\cos\omega t - (-\sin\omega t)\sin\omega t = \cos^2\omega t + \sin^2\omega t = 1$$

The columns are unit vectors ($\cos^2\omega t + \sin^2\omega t = 1$) and mutually orthogonal ($-\cos\omega t\sin\omega t + \sin\omega t\cos\omega t = 0$), so $[{}_0R_1][{}_0R_1]^T = I$.

Motion of S_1: $\boldsymbol{\omega}_{10} = \omega\mathbf{k}_1$, $\boldsymbol{\alpha}_{10} = \mathbf{0}$ (ω constant and $\mathbf{k}_1$ fixed), $\mathbf{r}_0^O = \mathbf{v}_0^O = \mathbf{a}_0^O = \mathbf{0}$. Poisson check: $\frac{d\mathbf{i}_1}{dt}\big|_0 = -\omega\sin\omega t\,\mathbf{i}_0 + \omega\cos\omega t\,\mathbf{j}_0 = \omega\,\mathbf{j}_1 = \omega\,\mathbf{k}_1 \times \mathbf{i}_1$ (chain rule).

Position vector. To reach P we go along the axis to the cross-section that contains her (height z), and then radially out to the floor, a distance R:

$$\mathbf{r}_1^P = z\,\mathbf{k}_1 + R\,\mathbf{e}_r$$

In the cross-section, $R\,\mathbf{e}_r$ is the hypotenuse (length R) of the shaded right triangle: the leg adjacent to φ lies along $\mathbf{i}_1$ and measures $R\cos\varphi$, the leg opposite to φ lies along $\mathbf{j}_1$ and measures $R\sin\varphi$. So, with φ the usual polar angle measured from $\mathbf{i}_1$:

$$\mathbf{e}_r = \cos\varphi\,\mathbf{i}_1 + \sin\varphi\,\mathbf{j}_1, \qquad \mathbf{e}_\theta = \frac{d\mathbf{e}_r}{d\varphi} = -\sin\varphi\,\mathbf{i}_1 + \cos\varphi\,\mathbf{j}_1$$

($\frac{d}{d\varphi}\cos\varphi = -\sin\varphi$, $\frac{d}{d\varphi}\sin\varphi = \cos\varphi$; $|\mathbf{e}_\theta| = \sqrt{\sin^2\varphi + \cos^2\varphi} = 1$, and $\mathbf{e}_\theta$ points in the sense of increasing φ, the sense of rotation). The matrix with columns $\mathbf{e}_r, \mathbf{e}_\theta, \mathbf{k}_1$ in $\mathcal{B}_1$ is a simple rotation by φ (matrix $\begin{pmatrix} \cos\varphi & -\sin\varphi & 0 \\ \sin\varphi & \cos\varphi & 0 \\ 0 & 0 & 1 \end{pmatrix}$, $\det = \cos^2\varphi + \sin^2\varphi = 1$; RR^T has diagonal entries $\cos^2\varphi + \sin^2\varphi = 1$ (and 1) and off-diagonal entries $\cos\varphi\sin\varphi - \sin\varphi\cos\varphi = 0$, so $RR^T = I$).

Derivatives and cross products used later. In S_1, by the chain rule, $\frac{d\mathbf{e}_r}{dt}\big|_1 = \frac{d\mathbf{e}_r}{d\varphi}\dot\varphi = \dot\varphi\,\mathbf{e}_\theta$ and $\frac{d\mathbf{e}_\theta}{dt}\big|_1 = (-\cos\varphi\,\mathbf{i}_1 - \sin\varphi\,\mathbf{j}_1)\,\dot\varphi = -\dot\varphi\,\mathbf{e}_r$. With $\mathbf{k}_1 \times \mathbf{i}_1 = \mathbf{j}_1$ and $\mathbf{k}_1 \times \mathbf{j}_1 = -\mathbf{i}_1$: $\mathbf{k}_1 \times \mathbf{e}_r = \cos\varphi\,\mathbf{j}_1 - \sin\varphi\,\mathbf{i}_1 = \mathbf{e}_\theta$ and $\mathbf{k}_1 \times \mathbf{e}_\theta = -\sin\varphi\,\mathbf{j}_1 - \cos\varphi\,\mathbf{i}_1 = -\mathbf{e}_r$. The part of $\mathbf{r}_1^P$ normal to $\boldsymbol{\omega}_{10}$ is $\mathbf{r}_\perp = R\,\mathbf{e}_r$ (the axial part $z\,\mathbf{k}_1$ is parallel to it).

Kinematics

In part (a) the astronaut is at rest on the floor, so $\mathbf{v}_1^P = \mathbf{0}$ and $\mathbf{a}_1^P = \mathbf{0}$. In part (b) she runs with a constant relative velocity:

Relative kinematics. $\mathbf{r}_1^P = R\,\mathbf{e}_r + z\,\mathbf{k}_1$ with $R\dot{\varphi} = u_\theta$, $\dot{z} = u_z$ (both constant). Differentiating in S_1 with the product and chain rules (R constant, $\mathbf{k}_1$ constant in S_1):

$$\mathbf{v}_1^P = R\dot{\varphi}\,\mathbf{e}_\theta + \dot{z}\,\mathbf{k}_1 = u_\theta\,\mathbf{e}_\theta + u_z\,\mathbf{k}_1$$

$$\mathbf{a}_1^P = \dot{u}_\theta\,\mathbf{e}_\theta + u_\theta\,(-\dot{\varphi}\,\mathbf{e}_r) + \dot{u}_z\,\mathbf{k}_1 = -\frac{u_\theta^2}{R}\,\mathbf{e}_r$$

($\dot{u}_\theta = \dot{u}_z = 0$ and $\dot{\varphi} = u_\theta/R$; the relative path is a helix of radius R; in S_1 she needs a centripetal acceleration u_θ^2/R).

Forces and Equations of Motion

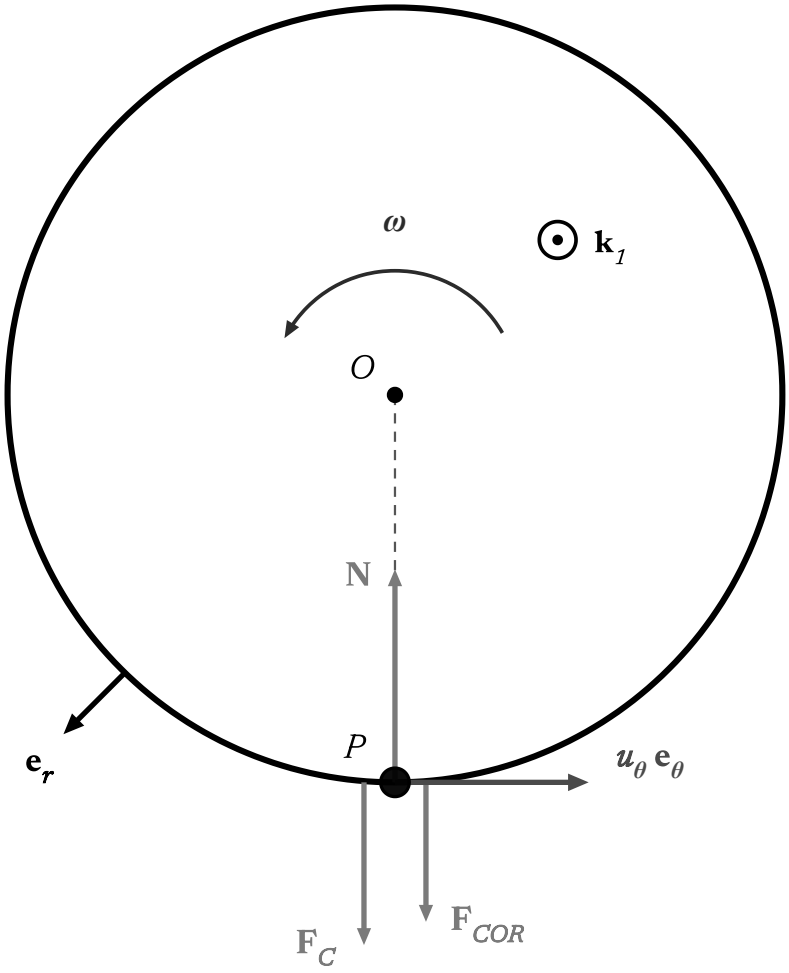


Figure 2. Forces in S_1 on the astronaut at P ($\mathbf{k}_1$ out of the paper). With $u_\theta > 0$ (running in the sense of rotation) both $\mathbf{F}_C = m\omega^2 R\,\mathbf{e}_r$ and $\mathbf{F}_{COR} = 2m\omega u_\theta\,\mathbf{e}_r$ point away from the axis; the floor force $\mathbf{N}$ points to the axis and balances them plus $m\mathbf{a}_1^P$.

The only actual force on the astronaut is the floor force $\mathbf{N}$ (no gravity in deep space). The inertia forces of Notes Eq. 4.62 are $\mathbf{F}_A = \mathbf{0}$ (fixed origin) and $\mathbf{F}_E = \mathbf{0}$ ($\boldsymbol{\alpha}_{10} = \mathbf{0}$); $\mathbf{F}_C = m\omega^2\mathbf{r}_\perp$ (Slide 30) and $\mathbf{F}_{COR} = -2m\boldsymbol{\omega}_{10} \times \mathbf{v}_1^P$ (Slide 31).

Centrifugal force. With $\mathbf{r}_\perp = R\,\mathbf{e}_r$, $\mathbf{k}_1 \times \mathbf{e}_r = \mathbf{e}_\theta$ and $\mathbf{k}_1 \times \mathbf{e}_\theta = -\mathbf{e}_r$:

$$\mathbf{F}_C = -m\,\omega\mathbf{k}_1 \times (\omega\mathbf{k}_1 \times R\,\mathbf{e}_r) = -m\,\omega\mathbf{k}_1 \times (R\omega\,\mathbf{e}_\theta) = m\omega^2 R\,\mathbf{e}_r$$

Coriolis force (running case), with $\mathbf{k}_1 \times \mathbf{e}_\theta = -\mathbf{e}_r$ and $\mathbf{k}_1 \times \mathbf{k}_1 = \mathbf{0}$:

$$\mathbf{F}_{COR} = -2m\,\boldsymbol{\omega}\mathbf{k}_1 \times (u_\theta\,\mathbf{e}_\theta + u_z\,\mathbf{k}_1) = 2m\omega u_\theta\,\mathbf{e}_r$$

Newton's law in S_1 (Notes Eq. 4.63), $m\mathbf{a}_1^P = \mathbf{N} + \mathbf{F}_C + \mathbf{F}_{COR}$. The unknown is the floor force $\mathbf{N}$ (the relative motion is prescribed, so no coordinate is unknown):

$$-m\frac{u_\theta^2}{R}\,\mathbf{e}_r = \mathbf{N} + m\omega^2 R\,\mathbf{e}_r + 2m\omega u_\theta\,\mathbf{e}_r$$

$$\mathbf{N} = -m\left(\omega^2 R + 2\omega u_\theta + \frac{u_\theta^2}{R}\right)\mathbf{e}_r = -\frac{m}{R}\left(\omega R + u_\theta\right)^2\mathbf{e}_r$$

(the bracket is the square $(\omega R + u_\theta)^2 = \omega^2 R^2 + 2\omega R u_\theta + u_\theta^2$ divided by R). At rest ($u_\theta = u_z = 0$) it reduces to the static balance $m\,\mathbf{0} = \mathbf{N} + m\omega^2 R\,\mathbf{e}_r$, $\mathbf{N} = -m\omega^2 R\,\mathbf{e}_r$.

Answers and Checks

(a) Spin rate for a given artificial gravity

> **Pedagogical Justification:** an astronaut at rest on the floor has $\mathbf{v}_1^P = \mathbf{0}$ and $\mathbf{a}_1^P = \mathbf{0}$, so Notes Eq. 4.63 becomes a static balance in S_1 between the floor reaction and the centrifugal force (the Coriolis force vanishes because $\mathbf{v}_1^P = \mathbf{0}$).

From the static balance, $\mathbf{N} = -m\omega^2 R\,\mathbf{e}_r$ and

$$g_{\text{art}} = \frac{|\mathbf{N}|}{m} = \omega^2 R$$

The floor pushes her toward the axis, and she feels a "gravity" $\omega^2 R$ pointing away from the axis. Setting $g_{\text{art}} = g$:

$$\boxed{\omega = \sqrt{\frac{g}{R}}}$$

Numbers ($R = 10$ m, $g = 9.81$ m/s^2):

$$\omega = \sqrt{\frac{9.81\ \mathrm{m/s^2}}{10\ \mathrm{m}}} = \sqrt{0.981}\ \mathrm{s^{-1}} \approx 0.9905\ \mathrm{rad/s}, \qquad T_{\mathrm{rev}} = \frac{2\pi}{\omega} \approx 6.34\ \mathrm{s}, \qquad \frac{60\,\omega}{2\pi} \approx 9.46\ \mathrm{rpm}$$

and the floor moves at $\omega R \approx 9.90$ m/s in S_0.

(b) Artificial gravity felt while running

> **Pedagogical Justification:** now $\mathbf{v}_1^P \neq \mathbf{0}$, so the Coriolis force appears, and her relative path may be curved, so $\mathbf{a}_1^P \neq \mathbf{0}$. Both are computed in S_1 and inserted in Notes Eq. 4.63 to obtain $\mathbf{N}$. The statement does not fix the running direction, so we take a general constant relative velocity with circumferential part u_θ and axial part u_z (running "around" the hull is $u_z = 0$, $u_\theta = \pm u$).

The reaction $\mathbf{N} = -\frac{m}{R}(\omega R + u_\theta)^2\,\mathbf{e}_r$ is purely radial, so no friction is needed to keep the motion uniform. The artificial gravity she feels is

$$\boxed{g_{\mathrm{art}} = \frac{(\omega R + u_\theta)^2}{R} = g\left(1 + \frac{u_\theta}{\omega R}\right)^2 = g + 2\omega u_\theta + \frac{u_\theta^2}{R}}$$

using $\omega^2 R = g$ from part (a). Particular cases:

- running around the hull **in the sense of rotation** ($u_\theta = +u$): $g_{\mathrm{art}} = (\omega R + u)^2/R > g$ (she feels heavier);
- running **against the rotation** ($u_\theta = -u$): $g_{\mathrm{art}} = (\omega R - u)^2/R < g$ for $u < 2\omega R$ (she feels lighter);
- running **along the axis** ($u_\theta = 0$, $u_z = u$): $g_{\mathrm{art}} = g$ (the Coriolis force vanishes because $\mathbf{v}_1^P \parallel \boldsymbol{\omega}_{10}$).

Numbers ($R = 10$ m, $\omega = 0.9905$ rad/s): at $u = 2$ m/s, $g_{\text{art}} = 14.17$ m/s^2 ($1.44\,g$) with the rotation and 6.25 m/s^2 ($0.64\,g$) against it; at $u = 5$ m/s, 22.2 m/s^2 ($2.26\,g$) and 2.41 m/s^2 ($0.25\,g$).

(c) Condition for no artificial gravity

> **Pedagogical Justification:** $g_{\text{art}} = (\omega R + u_\theta)^2/R$ is a perfect square, so it vanishes only when its base vanishes.

$$g_{\text{art}} = 0 \iff \omega R + u_\theta = 0 \iff \boxed{u_\theta = -\omega R, \quad u_z = 0}$$

She must run around the hull, against the rotation, at the speed of the floor ωR (≈ 9.90 m/s for the numerical case, a sprinter's speed). Then, by Notes Eq. 3.26, her absolute velocity is $\mathbf{v}_0^P = \mathbf{v}_1^P + \boldsymbol{\omega}_{10} \times \mathbf{r}_1^P = -\omega R\,\mathbf{e}_\theta + \omega R\,\mathbf{e}_\theta = \mathbf{0}$: she is at rest in inertial space, floating, while the floor slides under her feet. With an axial component ($u_z \neq 0$) and $u = \omega R$ the condition cannot be met, since then $|u_\theta| < \omega R$.

Checks

- **Inertial reading (cross-check).** In S_0 the astronaut moves on a circle (or helix) of radius R with tangential speed $\omega R + u_\theta$, so the floor must supply the centripetal force $m(\omega R + u_\theta)^2/R$: the same result. The rotating observer splits it into centrifugal $m\omega^2 R$, Coriolis $2m\omega u_\theta$ and relative centripetal $m u_\theta^2/R$ parts.
- **Limits:** $u_\theta \to 0$ gives $g_{\text{art}} = g$; for $|u_\theta| \ll \omega R$, $g_{\text{art}} \approx g \pm 2\omega u$, a correction due to the Coriolis force alone; a larger hull ($R \to \infty$ at fixed g) needs a smaller ω and makes the Coriolis correction $2\omega u = 2u\sqrt{g/R}$ smaller, which is why large habitats are more comfortable.
- **Orthogonality:** $\mathbf{e}_r \cdot \mathbf{e}_\theta = -\cos\varphi\sin\varphi + \sin\varphi\cos\varphi = 0$.
- **Dimensions:** $[\sqrt{g/R}] = \sqrt{\mathrm{m\,s^{-2}/m}} = \mathrm{s^{-1}}$; $[\omega^2 R] = [2\omega u_\theta] = [u_\theta^2/R] = \mathrm{m\,s^{-2}}$; $[N] = \mathrm{kg\,m\,s^{-2}} = \mathrm{N}$.
- **Verification (sympy):** $\mathbf{a}_1^P$, $\mathbf{F}_C$, $\mathbf{F}_{COR}$ and $\mathbf{N}$ are recomputed; the inertial double derivative of $[{}_0R_1]\{\mathbf{r}_1^P\}$ equals the Notes Eq. 3.27 decomposition and gives the same $\mathbf{N}$; the numbers above are reproduced.
- **Check against the official key:** `Problems.pdf` has no entry for Problem 9 in its solution keys, so the results are verified by independent derivation (the inertial reading and the sympy check above).

Source: Problems.pdf (PDF page 8); no official key